

Production
Idea's

A ONE H.P. 3600 RPM D.C. MOTOR
NEED NOT BE ANY LARGER THAN THIS.

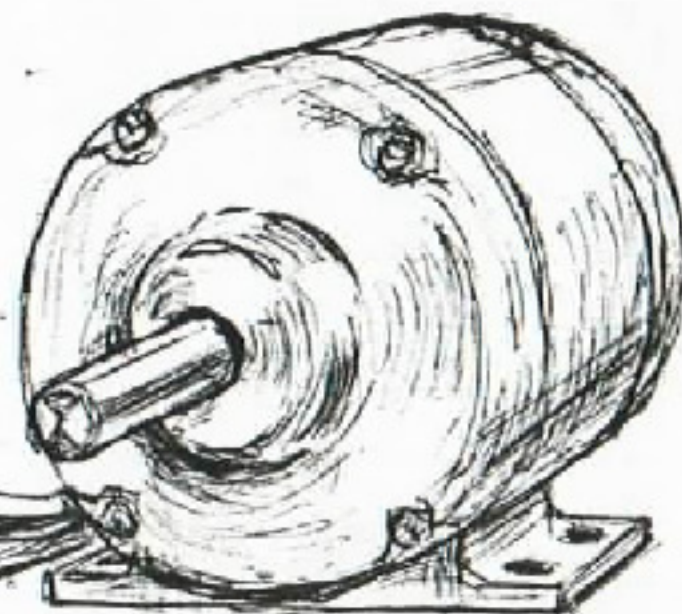
ACTUAL SIZE

SIZE:

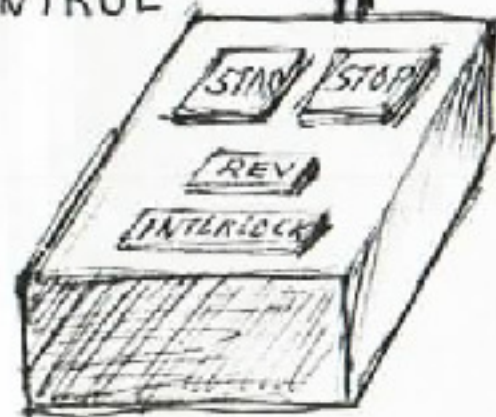
LENGTH: 2.75"
DIA: 2.25"
SHAFT DIA: .375"
WEIGH: 4.5 Lbs.

ELECTRICAL:

VOLTAGE: 10-20 V.DC.
10-24 V. DC.
STARTING SURGE: 9 A.
NO LOAD CUR: 50 MA.
1 HP LOAD CUR: 1.1 A @ 12 V.
(13 W.)
COGGING: 8% 1 HP LOAD
.1% NO LOAD
TORQUE: N.A.



REMOTE
ELECTRONIC
CONTROL



D.C. or
Rect A.C.

+
-

QUAD PATTERN BI-MONOPOLE ϕ VECTOR REPULSION FORCE D.C. MOTOR. BMRF MOTOR

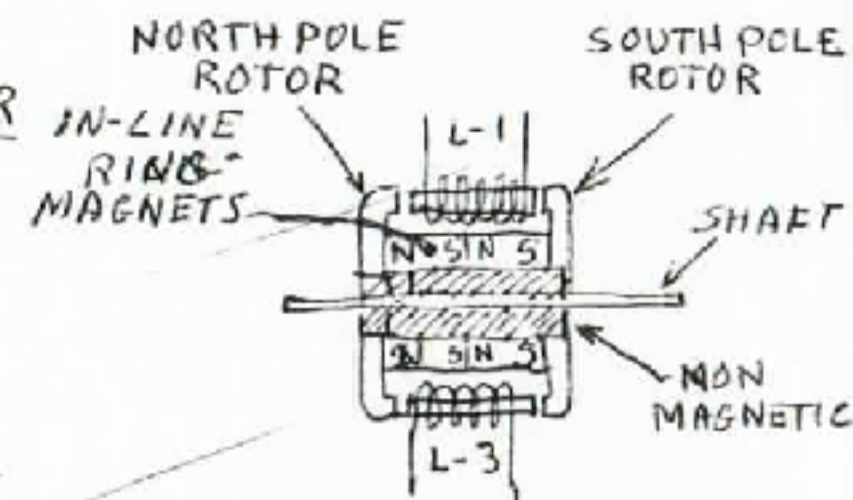
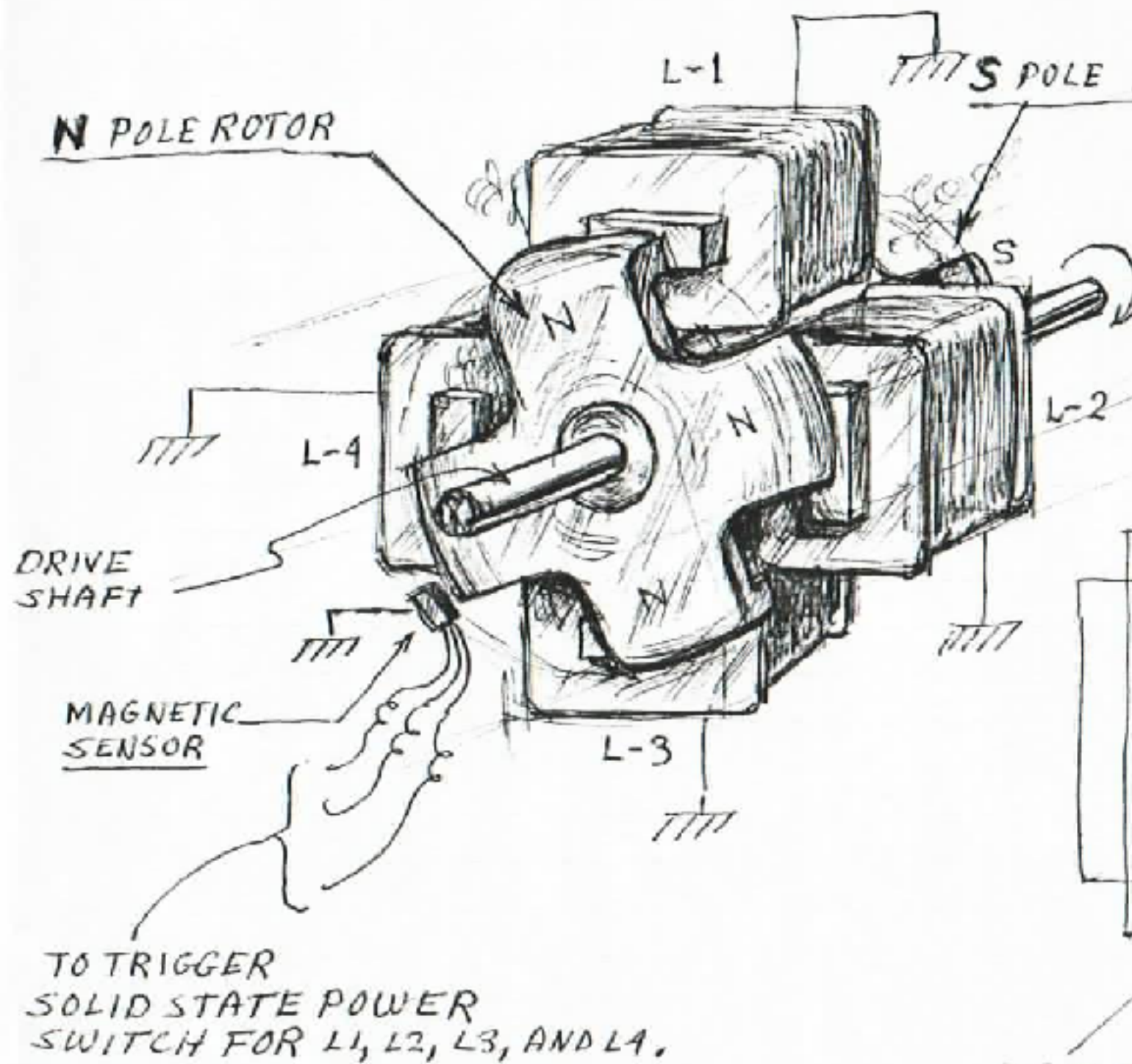
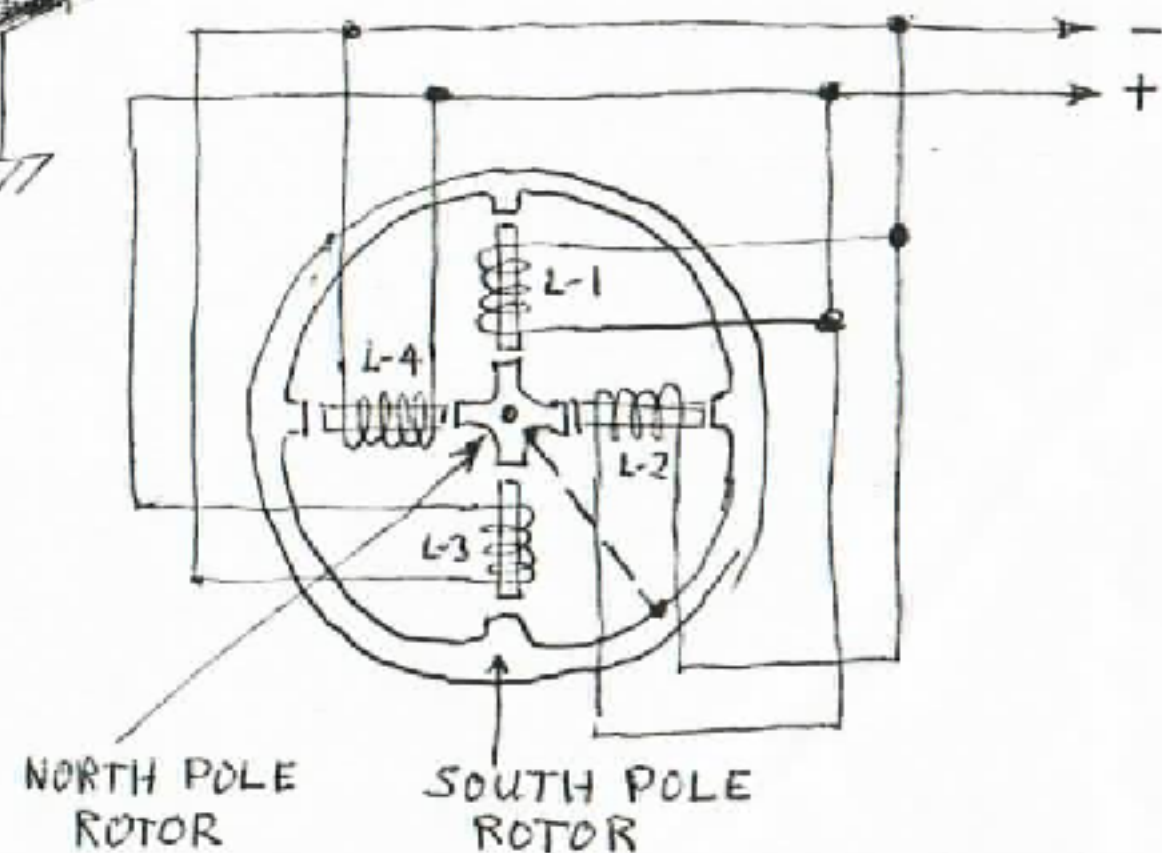


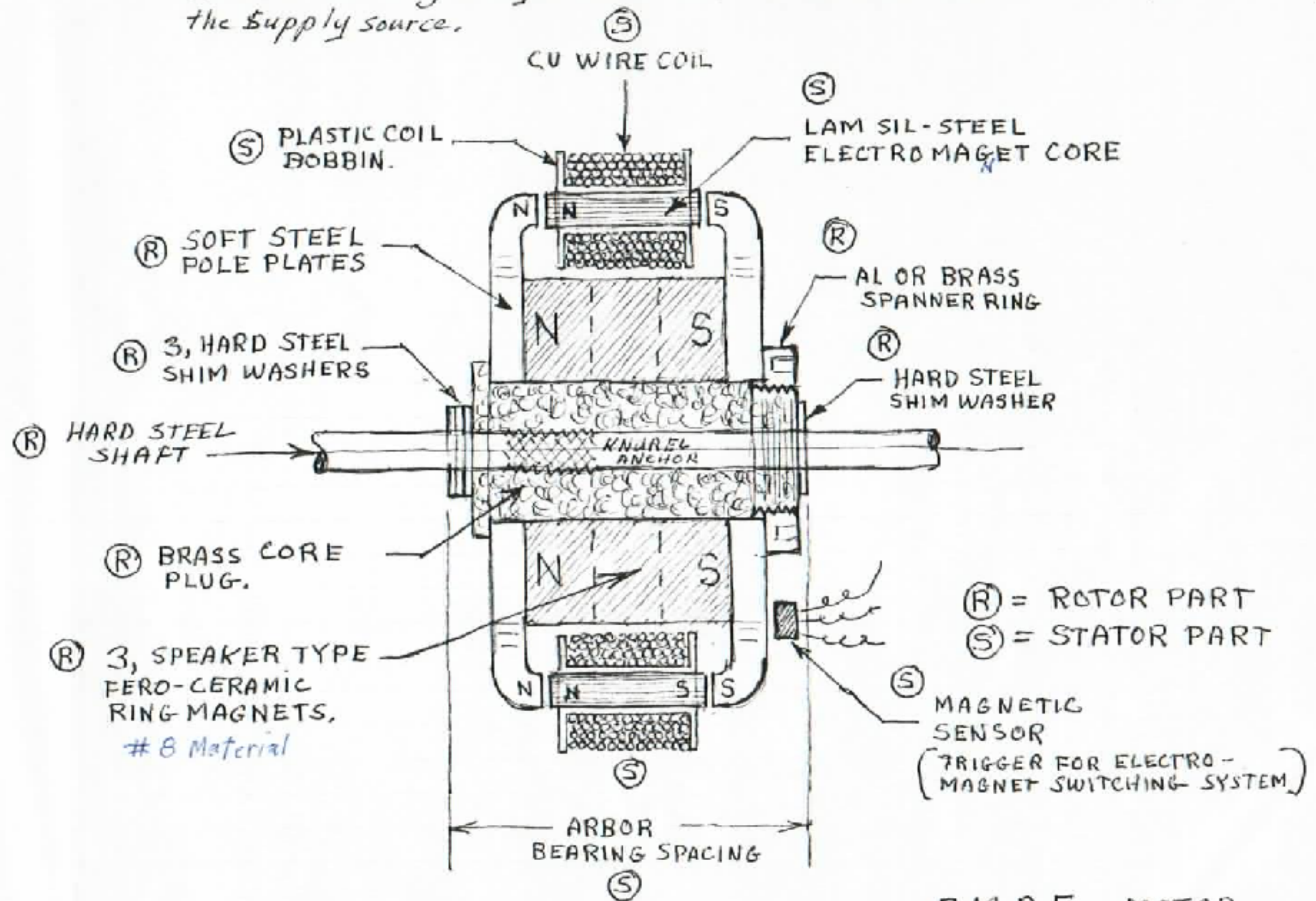
DIAGRAM:
LOW VOLTAGE, HIGH CUR.



B.E. COLE
11-29-85

ZERO VECTOR, BI-MONOPLE & REPULSION FORCE D.C. MOTOR

This motor is a zero hysteresis device that returns over 90% of motive energy to the supply source.

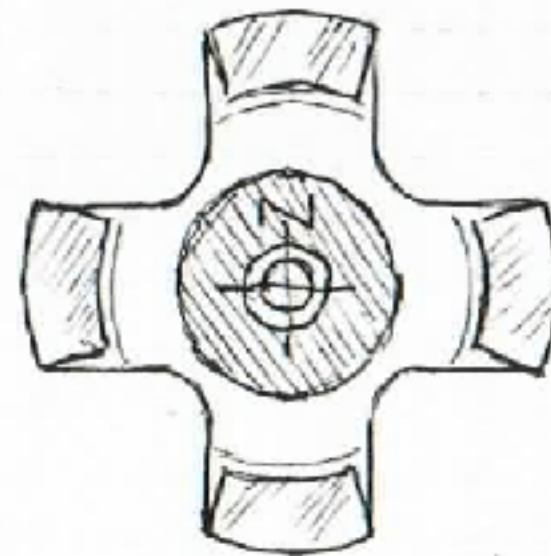
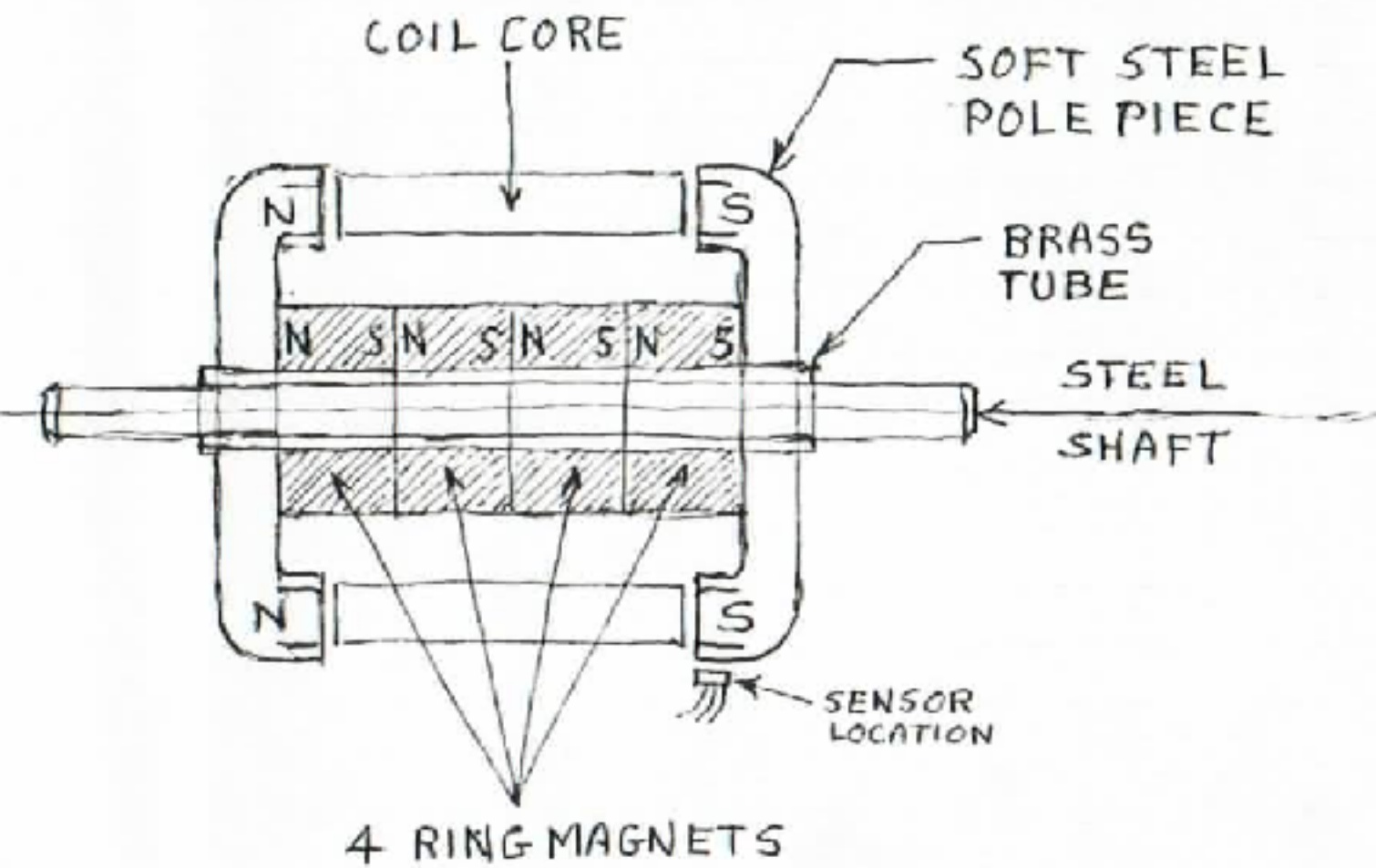


BMRF MOTOR

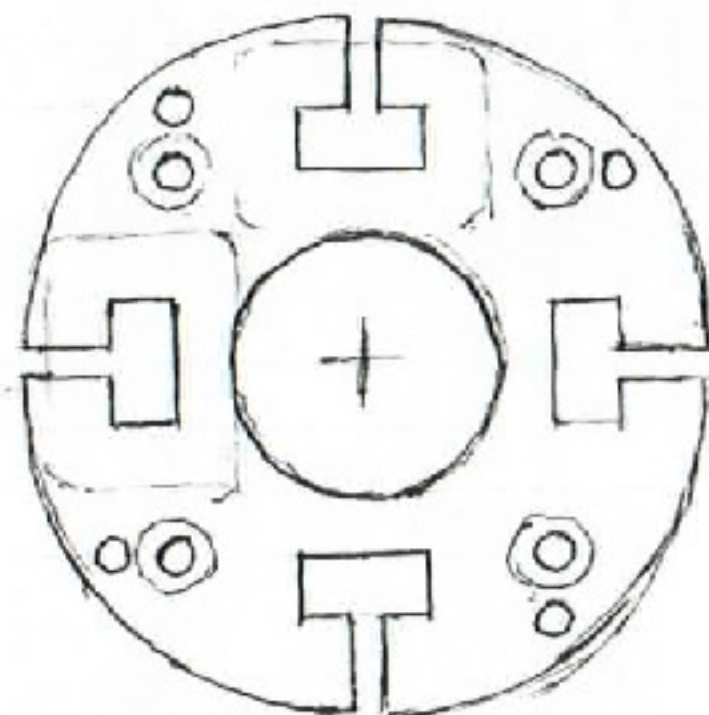
B.E. Ron Cole

11-18-85

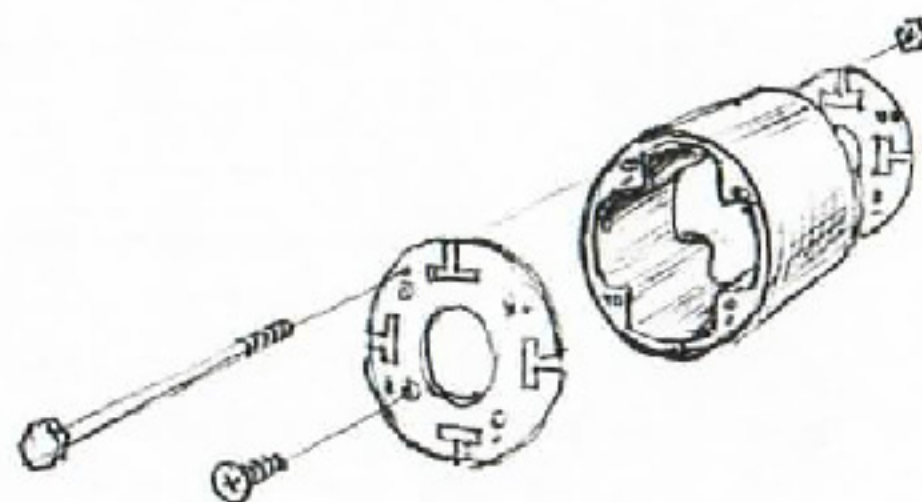
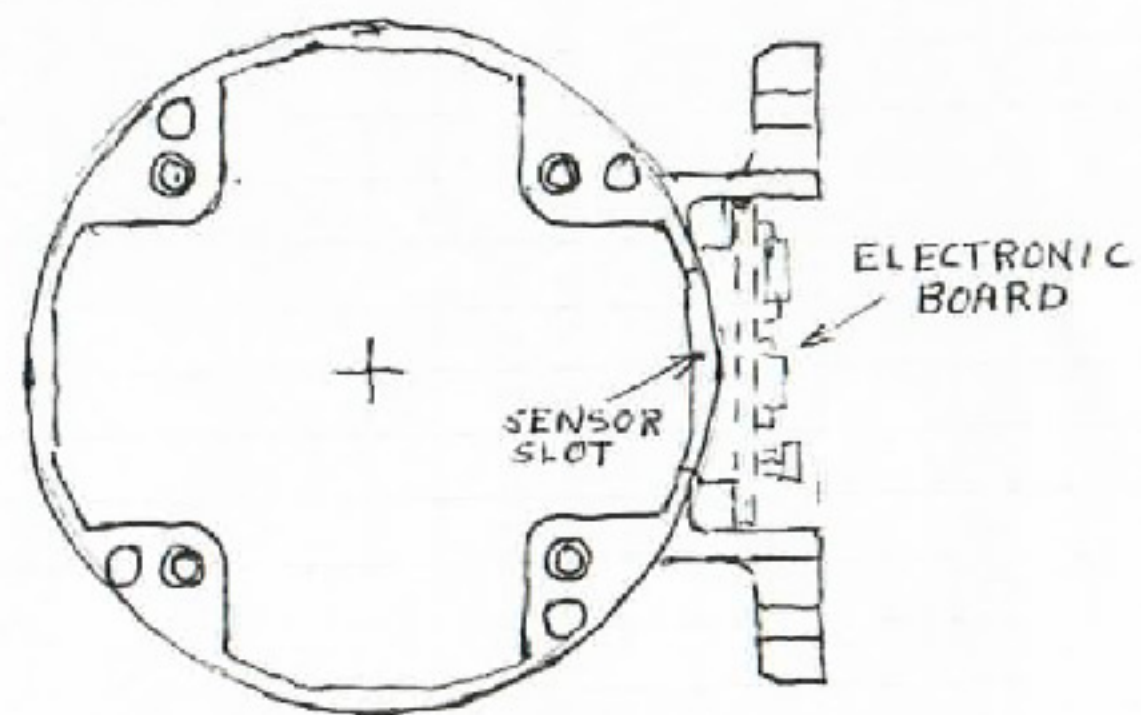
ROTOR ASSEMBLY



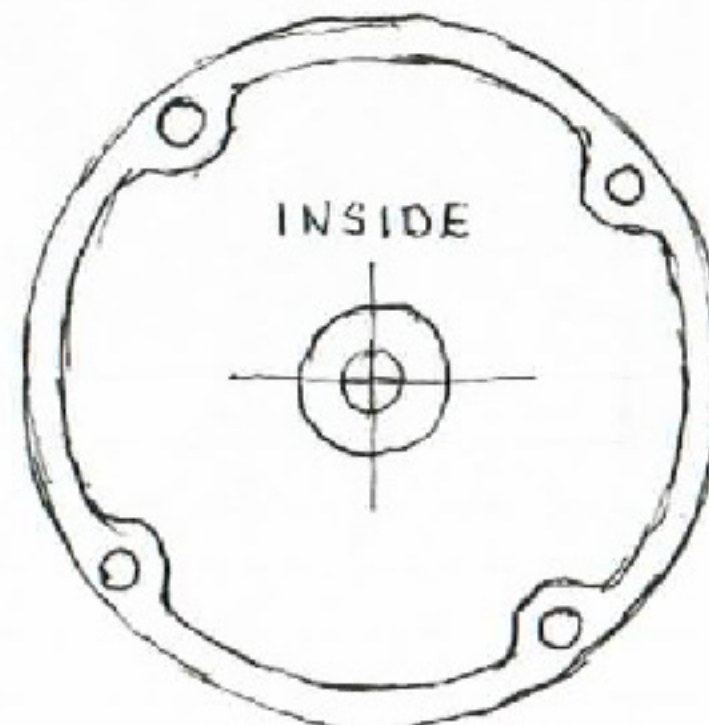
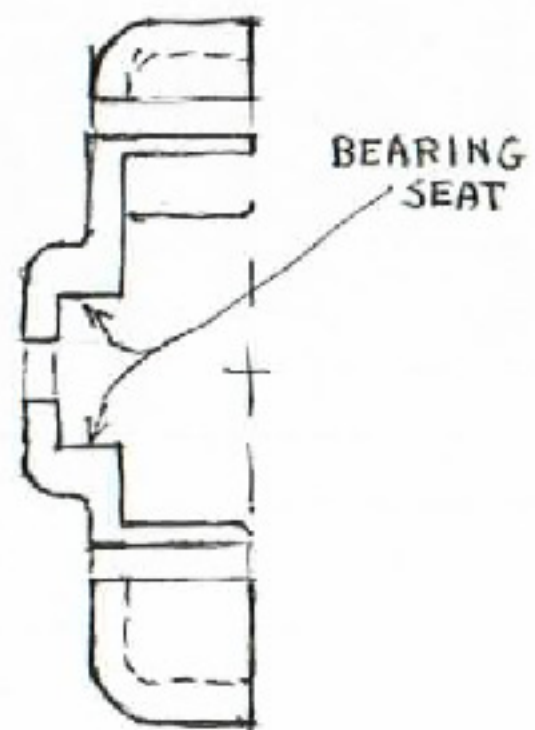
AL COIL MOUNTING
PLATE



PLASTIC SPACER
OUTTER SHELL



PLASTIC END BELL



- THIS MOTOR HAS OVER 6 TIMES THE OUTPUT TORQUE OF CONVENTIONAL MOTORS ~~BY~~ USING THE SAME AMOUNT OF COPPER COIL AREA.
- THE LAMINATED SILICON-STEEL CORE MATERIALS ARE ALMOST 13 TIMES LESS THAN THAT REQUIRED FOR AN EQUAL EQUIV. D.C. MOTOR.
- THE SOFT STEEL MAGNETIC PARTS ARE 6 TIMES GREATER THAN A CONVENTIONAL DESIGN.
- THE PERMINAT MAGNET MASS IS VIRTUALLY ON A 1 TO 1 RATIO OF THAT OF A CONVENTIONAL MOTOR.

MATERIALS COMPARISON:

CONVENTIONAL

COPPER = 1
 SIL-STEEL = 1
 SOFT STEEL = 1
 PERMINAT MAGNETS = 1

BMFR

$\frac{1}{6}$
 $\frac{1}{12}$
 6
 1

COST

Expensive
 Expensive
 Low Cost
 Expensive